5.3 Solving Systems of Linear Equations by Elimination

Name _____ Period ____ Date _____

Algebra I 5.3 · Student sheet

I can…

- solve systems by adding or subtracting equations.
- solve systems by multiplying first, then eliminating.
- solve real-life problems using elimination.

Part 1 · Make it cancel

System	Add, subtract, or multiply first? (by what?)	Solution
x + y = 6; x − y = 2		
3x + 2y = 7; 3x − y = 1		
x + 2y = 5; 3x − y = 1		
2x + 3y = 8; 5x − 2y = 1		

Part 2 · Notes: fill in the blanks

Add when the coefficients are _____. Subtract when they are _____.

If no coefficients match, _____ one or both equations first.

Multiply _____ term in the equation.

Part 3 · Examples (work them with the class)

Example 1

Solve.

3x + 2y = 4

5x − 2y = 12

Example 2

Solve.

2x + 3y = 1

x + 2y = 0

Example 3

Solve.

$4x + 3y = 10$

$3x - 2y = -1$

Example 4

2 pizzas and 3 salads cost $38. 4 pizzas and 1 salad cost $56. Find the cost of each.

Part 4 · Partner practice

Partner 1

Solve: $2x + y = 9$ and $2x - 3y = 1$.

Partner 2

Solve: $x + 3y = 7$ and $2x - y = 0$.